Chapter 36, Electric Fields Part A – Uniform Electric fields and Point Charges
Calculus Physics II

Fields (or Force Fields) - In physics a Field exists in a region of space if an object placed in that region experiences a force due to some quantity that object processes (mass and charge). The field can exist in empty space and the field creates non-contact forces, such as the force of the sun on the earth or an apple falling to earth (gravitational field).

- With non-contact forces the Force acts through its field such as the Gravitation Field or as you will learn the Electric Field for charged objects.
- Fields are real things, fields have energy, fields are energy!
- A condition in space that creates a force on a net charge, the condition is created by a net charge.

Two objects with mass will experience a force due to the gravitational field, the force is proposal to the mass of the two objects and the inverse square of the distance between. The gravitational field strength ($\vec{g}$) for a points mass is created by the mass m_1 and is given below. The force created by the field divided by the mass that creates the field gives the force exerted at that point in space. The gravitational force and field equations are given below.

$$\vec{\mathbf{F}}_{\mathbf{G}} = \left[G\frac{m_1 m_2}{r^2}\right]\hat{\mathbf{r}} \qquad \vec{\mathbf{g}} = \left[G\frac{m_1}{r^2}\right]\hat{\mathbf{r}} \qquad \vec{\mathbf{g}} = \frac{\vec{\mathbf{F}}_{\mathbf{g}}}{m} \qquad Units\ \frac{N}{kg}$$

Two objects with a net charge will experience a force due to the Electric field, the force is proposal to the net charge of the two objects and the inverse square of the distance between. The Electric field strength ($\vec{E}$) for a points charge is created by q_1 and is given below and is the force created by the field divided by the net charge that creates the field. The electrostatic force and field.

$$\vec{\mathbf{F}}_{\mathbf{E}} = \left[k\frac{q_1 q_2}{r^2}\right]\hat{\mathbf{r}} \qquad \vec{\mathbf{E}} = \left[k\frac{q_1}{r^2}\right]\hat{\mathbf{r}} \qquad \vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}_{\mathbf{E}}}{q} \qquad Units\ \frac{N}{C}$$

- The form of the field equations are similar, the force divided by the quantity that creates the field.

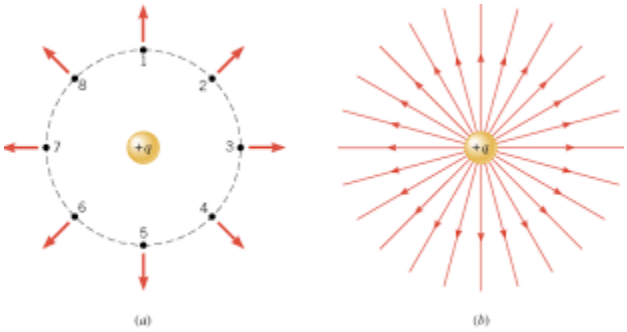
Electric Field ($\vec{E}$) - An electric field is said to exist at a point in space so that when an object with a net charge is placed at that point in space the object experiences an electrical force. The electric field is how charged particles interact with each other. The field is created by the charges tand exerts either attractive or repulsive forces on other charges within its vicinity in the electric field. The Electric Field is a vector because it creates a force and direction.

Units of electric field are **Newton per coulomb (N/C)**

Test Charge – A test charge is defined as an imagery positive charge that can be placed in an electric field to determine the properties of this field, the test charge is small enough so that it has no effect on the environment that it is placed in.

Test Charge is defined to be Positive and defines the electric field direction.

- An electric field is said to exist at any point in space when a test charge, placed at that point, experiences an electrical force.
- The direction of the electric field at a point is the same as the direction of the force experienced by a positive charge, for a positive charge the force is in the same direction as the electric field.
- The electric field points away from a positive charge because a positive test charge placed in the field created by the positive charge is repelled.
- The electric field points towards the negative charge because the positive test charge is attracted to the negative charge.

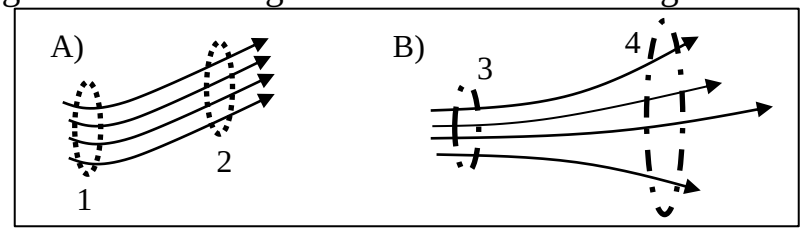


Electric field lines - Field lines are a means of describing an electric field, by using imaginary lines (or lines of force) to indicate the magnitude and direction of the field.

- The direction of an electric field line at any point is the direction of the force on the positive test charge and in the direction the charge would move if placed there.
- Field lines that are close together indicate that the field is strong and farther apart the field is weak.
- The number of field lines through an area is proportional to the strength/magnitude of the electric field.
- Electric field lines begin on a positive charge and end on a negative charge or go to infinity; they do not start or stop in mid-space.
- Field lines never cross.
- Electric field lines are perpendicular to the surface they emerge from or terminate into.

A) The electric field lines are the same spacing therefore the electric field strength is the same from position 1 to 2.

B) The electric field lines get farther apart from position 3 to 4 therefore the electric field gets weaker. Region 3 has a stronger electric field than in region 4.



<u>Superposition of Electric Field – The net Electric Field ($\vec{\mathbf{E}}_{\mathbf{net}}$)</u>- If there is more than one source of an electric field at a point in space then the net field is the vector sum of all the individual field present at that point giving the resultant field at that point in space.

The net electric field is the vector sum of all electric fields at a particular point in space

$$\boldsymbol{\Sigma\vec{E}} = \vec{\mathbf{E}}_{\mathbf{net}} = \vec{\mathbf{E}}_{\mathbf{1}} + \vec{\mathbf{E}}_{\mathbf{2}} + \vec{\mathbf{E}}_{\mathbf{3}} \ldots$$

If there are two Electric fields at a point in space are given by

$$\vec{\mathbf{E}}_{\mathbf{1}} = (E_{1x})\hat{\mathbf{x}} + (E_{1y})\hat{\mathbf{y}} \qquad \vec{\mathbf{E}}_{\mathbf{2}} = (E_{2x})\hat{\mathbf{x}} + (E_{2y})\hat{\mathbf{y}}$$

Taking the dot product with the unit vector $\hat{x}$ to find the net force in the x-direction

$$\boldsymbol{\Sigma\vec{E}} \bullet \hat{\mathbf{x}} = (\vec{\mathbf{E}}_{\mathbf{1}} + \vec{\mathbf{E}}_{\mathbf{2}}) \bullet \hat{\mathrm{x}} = \Sigma \mathrm{E_x}$$

$$\Sigma \mathrm{E_x} = E_{1x} + E_{2x}$$

Taking the dot product with the unit vector $\hat{y}$ to find the net force in the y-direction

$$\vec{\mathbf{E}}_{\mathbf{net}} \bullet \hat{\mathbf{y}} = (\vec{\mathbf{E}}_{\mathbf{1}} + \vec{\mathbf{E}}_{\mathbf{2}}) \bullet \hat{\mathrm{y}} = \Sigma \mathrm{E_y}$$

$$\Sigma \mathrm{E_y} = E_{1y} + E_{2y}$$

The net electric field is then equal to

$$\boldsymbol{\Sigma\vec{E}} = (\Sigma \mathrm{E_x})\hat{\boldsymbol{x}} + (\Sigma \mathrm{E_y})\hat{\boldsymbol{y}}$$

$$\vec{\mathbf{E}}_{\mathbf{net}} = \boldsymbol{\Sigma\vec{E}} = (\mathrm{E_{1x}} + \mathrm{E_{2x}})\hat{\mathbf{x}} + (\mathrm{E_{1y}} + \mathrm{E_{2y}})\hat{\mathbf{y}}$$

The magnitude and direction are given by

$$|\vec{\mathbf{E}}_{\mathbf{net}}| = \sqrt{\vec{E} \bullet \vec{E}} = \sqrt{E_x^2 + E_y^2}$$

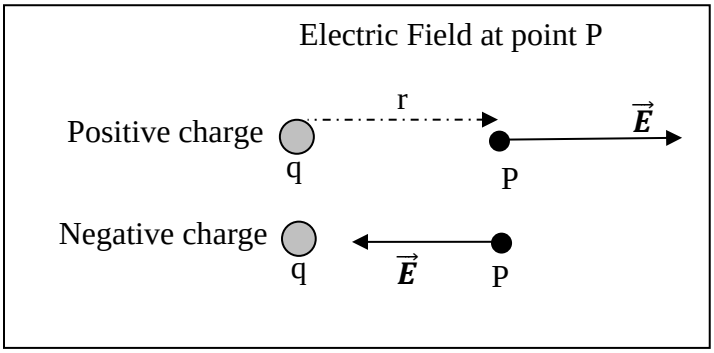
The direction of the net Electric field is given by

$$\alpha_{net} = cos^{-1}\left(\frac{E_x}{|\vec{E}|}\right) \qquad \beta_{net} = cos^{-1}\left(\frac{E_y}{|\vec{E}|}\right)$$

<u>Electric Field due to a point charge</u> – All charges produce an electric field, the electric field that is produced by the charge is the source of the force the charge exerts on other charges. The electric field $\vec{E}$ produced by a point charge q at a distance r from that charge is given as

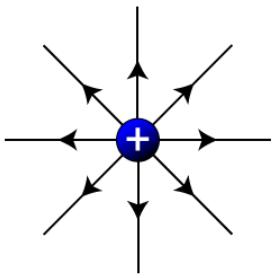
$$\vec{\mathbf{E}}_{\mathbf{point}} = \left[\frac{kq}{r^2}\right]\hat{\mathbf{r}}$$

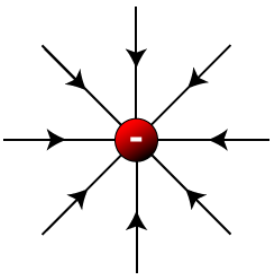
$\vec{\mathbf{E}}_{\mathbf{point}}$ – Electric Field Strength at point r from the source point charge (q).
$k = 9.0 \times 10^9\ Nm^2/C^2$ Electrostatic Constant $k = 1/4\pi\varepsilon_o$ $\varepsilon_o = 8.85 \times 10^{-12}\ C^2 / Nm^2$
q – net charge on the object that is creating the electric field.
r – straight line distance between object creating the charge and the point of interest.
$\hat{\mathbf{r}}$ – unit vector in the direction of the electric field determined by the Positive test charge.

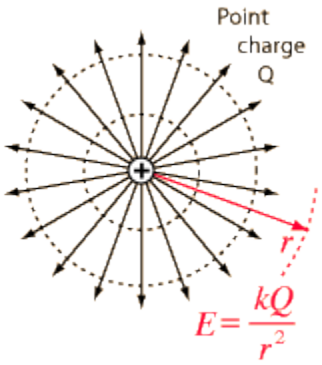


➢ q is the net charge producing the Electric field.

- ❖ For a Positive charge the electric field radiates outward from the charge and gets weaker.
- ❖ For a Negative charge the field points towards the charge and get weaker the farther from the charge.

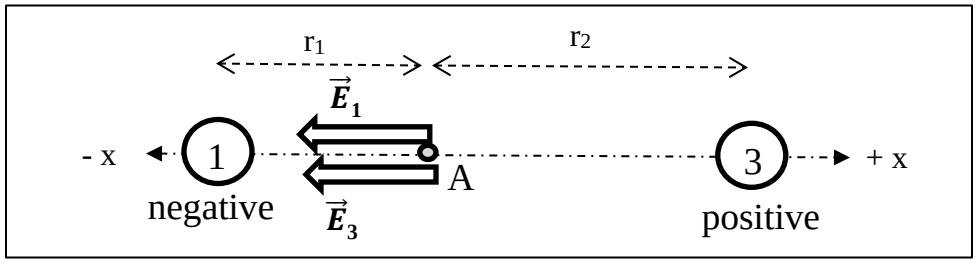






<u>Electric field due to multiple point charges along the same line</u> – for two or more point charges that are present, the net electric field created in space at point P is the vector sum of the field from each of the point charges. For point charges along the same line any point in space along that line is the sum of the individual fields taking into account their direction.

Example: If there are two balls with the net charge as shown, what is the magnitude and direction of the net electric field at point A, due to the charge on ball 1 (negative 2 μC) and ball 3 (positive 6 μC)? $r_1 = 0.1$ m and $r_2 = 0.2$ m, $k = 9.0 \times 10^9$ Nm^2/C^2 , 1 μC = 1×10^{-6} C (Assume the balls are not moving)



1) Only consider point A and the electric fields created at this point by balls 2 and 3 separately due to their charge.

   Note: There is no charged object or charge at point A, it is empty space only the electric field is being produced at this point!

2) Determine the direction of the electric field at point A.

   - Only consider point A a distance r from the charge that is creating the electric field, the electric field ($\vec{E}_1$) from ball 1 is pointed in the negative x direction because ball 1 is negatively charged and the electric field points towards the charge if it has a negative charge.
   - Now consider ball 3's effect at point A, ball 3 has a positive charge so the electric field ($\vec{E}_3$) points away from ball 3 which means that at point A the field from ball 3 is pointing in the negative x direction.

3) Calculate the magnitude of each of the forces using each pair's charge (q) and distance apart (r)

   $|\vec{E}_1| = kq_1/r_1^2 = (9.0 \times 10^9\ Nm^2/C^2)(2 \times 10^{-6} C)/(0.1m)^2 = 1.8 \times 10^6$ N/C

   $|\vec{E}_3| = kq_3/r_2^2 = (9.0 \times 10^9\ Nm^2/C^2)(6 \times 10^{-6} C)/(0.2m)^2 = 1.35 \times 10^6$ N/C

   Note: There is no sign (+ or -) associated with the magnitude of the field.

4) Sum the forces acting on the object in question and solve.

$\Sigma E_{x\ at\ P} = (-|\vec{E}_1|) + (-|\vec{E}_3|) = -1.8 \times 10^6$ N/C $+ (-1.35 \times 10^6$ N/C$) = 3.15 \times 10^6$ N/C

Pointing along the negative x-axis

5) If a third charged ball, let's call it ball 2 (negative 4 μC) is placed at point A, what force would it experience and in what direction?

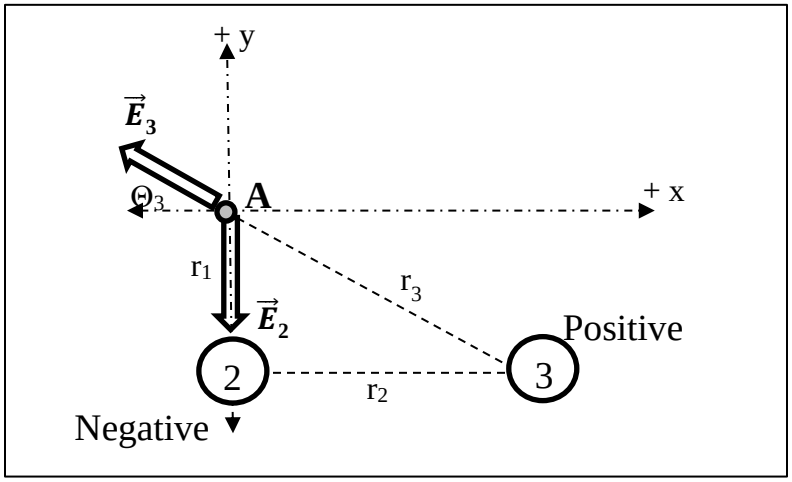
$|\vec{\mathbf{F}}| = q|\vec{E}| = (4 \times 10^{-6}\ C)(3.15 \times 10^{6}\ N/C) = 12.6\ N$ along the positive x-axis

Because the electric field points in the negative x direction then a positively charge ball would experience a force in the negative x direction if placed at point A but we have a negatively charged ball so it would expense a force in the opposite direction as the positive which would be in the positive x direction.

This is the same Electric force as the previous example (Chapter 35) for point charges and the force!

Electric Field due to multiple charges in a plane: If there are two balls with the net charge as shown, what is the magnitude and direction of the net electric field at point A due to balls 2 and 3?

The Charge on ball 2 (negative 4 μC) and ball 3 (positive 6 μC)? $r_1 = 0.1$ m and $r_2 = 0.2$ m, $r_3 = 0.224$ m, $\Theta_3 = 26.6°$ $k = 9.0 \times 10^9\ Nm^2/C^2$ (Assume the balls are not moving)



1) Only consider point A and the electric fields created there by balls 2 and 3 separately due to their charge. Note: There is no charged object or charge at point A, it is empty space only the electric field is being produced by the two net charges at this point!

2) Determine the direction of the electric field at point A.

   - Only consider point A with nothing there, the electric field ($\vec{E}_2$) from ball 2 is pointed in the negative-y direction because ball 2 is a negatively charged and the electric field points towards the charge if it has a negative charge.

- Now consider ball 3's effect at point A, ball 3 has a positive charge so the electric field ($\vec{E}_3$) points away from ball 3 which means that at point A the field from ball 3 is pointing at an angle $\Theta_3 = 26.6°$ above the negative x axis.

3) Calculate the magnitude of each of the forces using each pair's charge (q) and distance apart (r)

$$\left|\vec{\mathbf{E}}_2\right| = \frac{kq_2}{r_1^2} = \frac{(9.0 \text{ x } 10^9 \text{ N}m^2/C^2)(4\text{x}10^{-6}\text{C})}{(0.1\text{m})^2} = 3.6x10^6\ N/C$$

$$\left|\vec{\mathbf{E}}_3\right| = \frac{kq_3}{r_3^2} = \frac{(9.0 \text{ x } 10^9 \text{ N}m^2/C^2)(6\text{x}10^{-6}\text{C})}{(0.224\text{m})^2} = 1.08x10^6\ N/C$$

4) Determine the components of the vectors and put them in vector form.

$\vec{\mathbf{E}}_2 \bullet \hat{\mathbf{x}} = E_{2x} = \left|\vec{\mathbf{E}}_2\right|\cos(90°) = 0$ N/C
$\vec{\mathbf{E}}_2 \bullet \hat{\mathbf{y}} = E_{2y} = \left|\vec{\mathbf{E}}_2\right|\cos(180°) =$ - 3.6 x 10^6 N/C (along - y axis)
$\vec{\mathbf{E}}_3 \bullet \hat{\mathbf{x}} = E_{3x} = \left|\vec{\mathbf{E}}_3\right|\cos(180° - 26.6°) =$ - 9.66 x 10^5 N/C (x component along - x-axis)
$\vec{\mathbf{E}}_3 \bullet \hat{\mathbf{y}} = E_{3y} = \left|\vec{\mathbf{E}}_3\right|\cos(90° - 26.6°) =$ + 4.84 x 10^5 N/C (y component along + y-axis)

$$\vec{\mathbf{E}}_2 = (E_{2x})\hat{\boldsymbol{x}} + (\Sigma E_{2y})\hat{\boldsymbol{y}} \quad \text{and} \quad \vec{\mathbf{E}}_3 = (E_{3x})\hat{\boldsymbol{x}} + (\Sigma E_{3y})\hat{\boldsymbol{y}}$$

$\vec{\boldsymbol{E}}_2 = (0 \text{ N/C})\hat{\boldsymbol{x}} + (-3.6 \text{ x } 10^6 \text{ N/C})\hat{\boldsymbol{y}}$ and $\vec{\boldsymbol{E}}_3 = ((-9.66 \text{ x } 10^5 \text{ N/C})\hat{\boldsymbol{x}} + (+4.84 \text{ x } 10^5 \text{ N/C})\hat{\boldsymbol{y}}$

5) Sum the forces acting on the object and solve for the resultant vector ($\mathbf{E_{net}}$)

$$\Sigma\vec{\mathbf{E}} = (\Sigma E_x)\hat{\boldsymbol{x}} + (\Sigma E_y)\hat{\boldsymbol{y}}$$

$$\vec{\mathbf{E}}_{net} = (E_{1x} + E_{2x})\hat{\mathbf{x}} + (E_{1y} + E_{2y})\hat{\mathbf{y}}$$

The net force in components form is given as

$$\vec{\mathbf{E}} = (-9.66 \text{ x } 10^5 \text{ N/C})\hat{\mathbf{x}} + (-3.12 \text{ x} 10^6 \text{ N/C})\hat{\mathbf{y}}$$

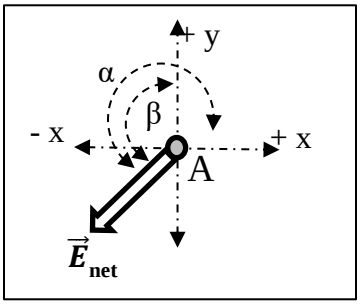
The magnitude and direction of the electric field at point A is

$$\left|\Sigma\vec{E}\right| = \left|\vec{E}_{net}\right| = \sqrt{E_x^2 + E_y^2} = 3.27 \text{ x } 10^6 N/C$$

The direction of the net Electric field is

$$\alpha_{net} = cos^{-1}\left(\frac{E_x}{\left|\Sigma\vec{E}\right|}\right) = 107° \qquad \beta_{net} = cos^{-1}\left(\frac{E_y}{\left|\Sigma\vec{E}\right|}\right) = 163°$$

Therefore, the magnitude of the electric field at point A due to the charge on ball 2 and 3 is 3.27 x 10^6 N/C and points in a direction that is 73° below the negative x-axis

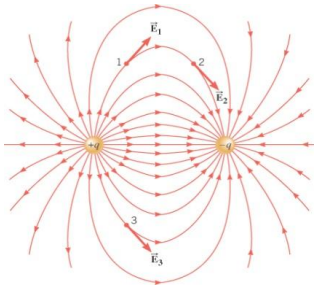


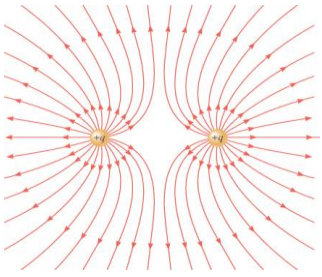
6) If a third ball, let's call it ball 1 (negative 2 μC) is placed at point A, what is the magnitude and direction of the force on ball 1?

$\left|\vec{F}\right|$= qE = (2 x 10^{-6} C)(3.27 x 10^6 N/C) = 6.5 N at an angle of 26.6° above the positive x-axis (because the ball have a negative charge it points in the opposite direction as the electric field direction)

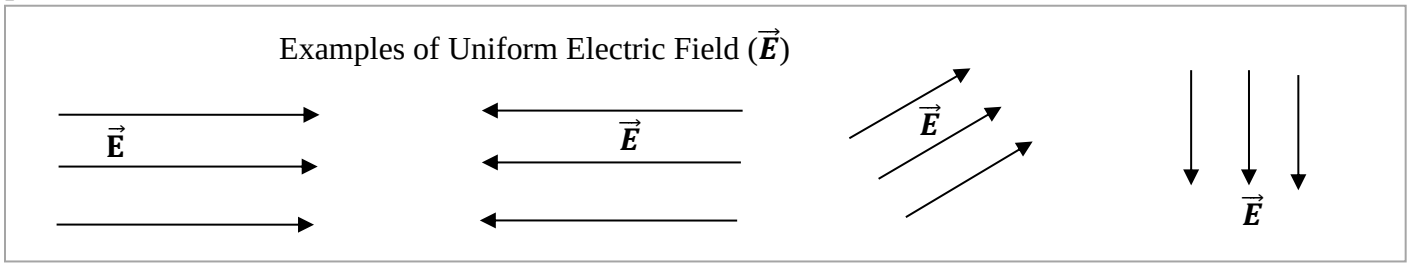
<u>Electric Dipole</u> – Two equal but oppositely type of charged object that is separated by some distance but are still bound to each other. Electric dipoles can be naturally occurring, or they can be induced by bringing a charge near a neutral molecule (polarization).

Electric Field lines for opposite charges (dipole) and like charges.





Uniform Electric Field – In a uniform electric field the electric field strength ($\vec{\mathbf{E}}$) does not change value with change in location or position. The electric field lines are equally spaced and parallel.



The Strength and Force in a Uniform Electric Field - The strength of an electric field ($\vec{E}$) at a point is equal to the force experienced by a unit positive test charge placed at that point. Because that electric field strength is a force per unit charge it is a vector and points in the direction of the force on the positive charge.

The Electric Field is given as the force per unit charge at a given location; the electric field can change over a region or can be constant throughout that region.

$\hat{\mathbf{E}}$ is the unit vector in the direction of the electric field

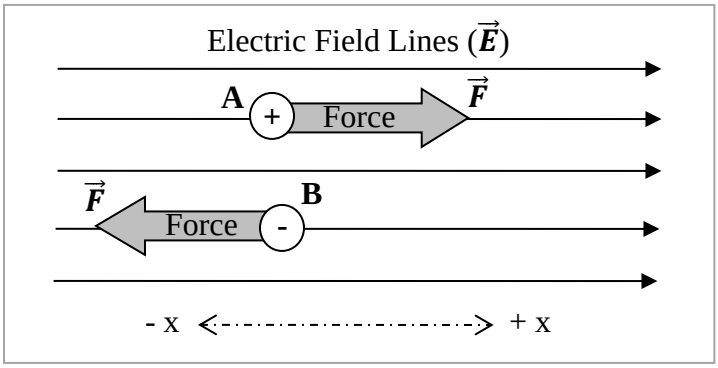
$$\vec{\mathbf{E}} = \left[\frac{Electrostatic\ Force}{Net\ Charge}\right]\hat{\mathbf{E}} = \left[\frac{|\vec{\mathbf{F}}_{\mathbf{E}}|}{q}\right]\hat{\mathbf{E}}$$

$$|\vec{\mathbf{E}}| = \frac{|\vec{\mathbf{F}}_{\mathbf{E}}|}{q}$$

If a charged object is placed at a point in space that has an electric field strength $\vec{E}$, Then that object will experience a force at that point along the electric field line.

$$|\vec{\mathbf{F}}_{\mathbf{E}}| = q|\vec{\mathbf{E}}|$$

If the net charge q on an object is positive, then the electric force is in the same direction as the field. If an object that has a net charge q is negative, then $\vec{F}_{\mathrm{E}}$ will be opposite in direction to $\vec{E}$.



Example: From the diagram above if Sphere A has a net positive charge of 10 μC and is placed in a uniform electric field with a field strength of 40,000 N/C pointed in the positive x direction, what force does Sphere A experience due to the electric field and in what direction.

$|\vec{F}| = q|\vec{E}|$ = (10 x 10^{-6} C)(40,000 N/C) = 0.4 N along positive x-axis

Because Sphere A has a positive charge the force is along the direction of the electric field which is in the positive x direction.

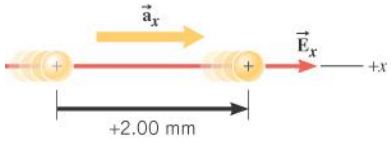
Sphere B has a net negative charge of 10 μC and is placed in a uniform electric field with a field strength of 40,000 N/C pointed in the positive x direction, what force does Sphere B experience due to the electric field and in what direction.

$|\vec{F}| = q|\vec{E}|$ == (10 x 10^{-6} C)(40,000 N/C) = 0.4 N along negative x-axis

Because Sphere B has a negative charge the force is in the opposite direction of the electric field, which is in the positive x direction, therefore the force on the negative charge is along the negative x direction.

Newton's Laws and The Electrostatic Force - The amazing thing about Newton's Laws is that it does not matter the origin of the force, the results of the net force have the same consequences, $\Sigma\vec{F} = m\vec{a}$ still applies. For a charged particle under the influence of only the electrostatic force in a uniform electric field $\Sigma\vec{F} = m\vec{a} = q\vec{E}$ and all of Newton's laws of motion and the kinematic equations apply.

$$\mathbf{\Sigma\vec{F} = \vec{F}_E = q\vec{E} = m\vec{a}}$$



- ✓ Newton's Laws apply to Electrostatic forces just as they did to applied forces on objects – "a force is a force"
- ✓ The kinematic equations apply to the motion of charged particles as long as the conditions are met that the

1) Motion is all along the same line
2) The acceleration is constant in a uniform electric field.

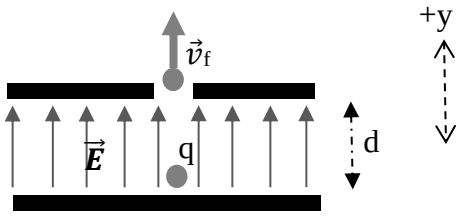
Question: Which would have the larger acceleration in the same electric field an electron or a proton? (Both have the same magnitude of charge but the mass of the proton is about 2000 time that of the electron so the electron would have the largest acceleration)

Example: If a sphere with a mass of m = 5 x10^{-3} kg with a net positive charge of q = 15 μC is placed in a uniform electric field ($|\vec{E}|$ = 500 N/C) as shown, if the sphere starts from rest, what is the velocity of the sphere when it exits the field (d = 0.25 m)? (Ignore gravity)

Summing the forces acting on the sphere and knowing that $\vec{F}_e = q\vec{E}$ in a uniform electric field

$\Sigma F_{y\ sphere} = F_e = qE = ma_y$ therefore

$a_y = qE/m = 1.5\ m/s^2$



Using the kinematic equations and the given information

$v_i = 0$ m/s
$v_f = ?$
$a = 1.5\ m/s^2$
$\Delta x = d = 0.25$ m

$v_f^2 = v_i^2 + 2a\Delta x$ solving for the final velocity

$v_f^2 = 2a\Delta x$

$v_f = (2a\Delta x)^{1/2} = 0.87$ m/s

Example: A negatively charged ball (m_{ball} = 5 x 10^{-3} kg, q_{ball} = 10 μC) enters a uniform electric field with a magnitude of $|\vec{E}|$ = 5,000 N/C along the field lines traveling in the + x direction, the ball comes to a stop in the electric field 2 seconds after entering it.

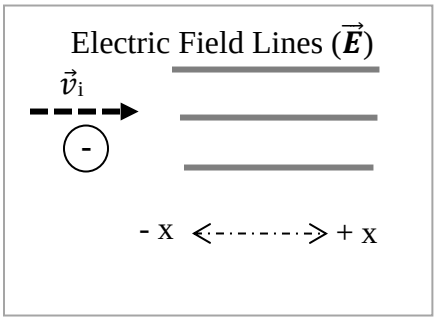
A) Determine the initial velocity of the ball before entering the electric field?

$v_i = ??$
$v_f = 0$ m/s
$a = ?$
$\Delta t = 2$ s

The only force is the electrostatic force (F_E) therefore

$\Sigma F_x = -F_E = ma_x$ solving for a_x

$a_x = -F_E/m = qE/m = -10\ m/s^2$



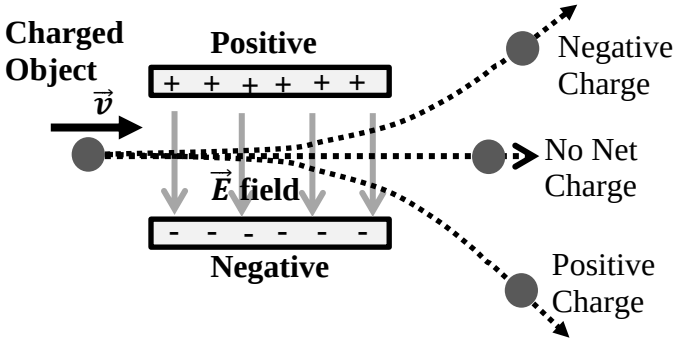
Knowing the acceleration $v_f = v_i + at$ can be used solving for v_i

$v_i = at = (-10\ m/s^2)(2s) = 20$ m/s

B) Determine the direction of the Electric Field?

In order to stop the negative ball, the electric force acting on it must be in the opposite direction as the velocity therefore the force acting the ball is in the negative x direction. The ball has a negative charge therefore the electric field must be in the pointing in the negative x direction.

<u>Electrostatic Force on a charged object moving perpendicular to the Electric Field</u> - When the electric field ($\vec{\mathbf{E}}$) is perpendicular to the direction of the velocity ($\vec{v}$) of the charged particle there is a change in direction of the particle but with no change in speed.



- Positive charge goes down because the force is in the same direction as the electric field which points down
- Negative charge goes up because the force is in the opposite direction as the electric field which points down
- A neutral particle will not be affected because there is no net charge therefore no force
- The larger the charge on the particle the larger the force therefore the larger the deflection
- Once the charged particle leaves the region with the electric field there is no more defection, and it moves in a straight line at a constant velocity.